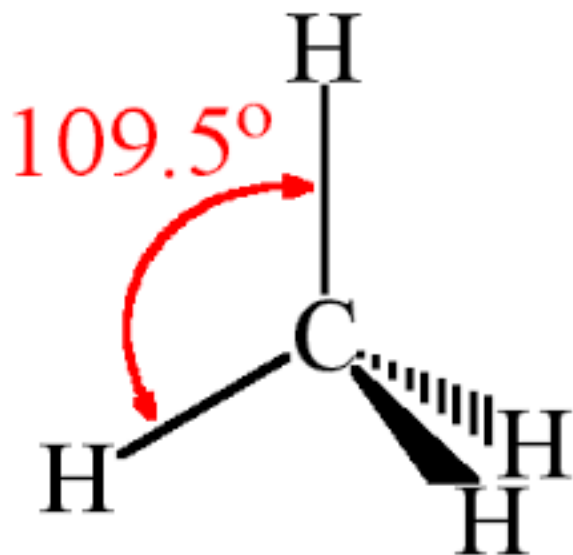


Lecture 01

STEREOCHEMISTRY

Stereochemistry –

three-dimensional arrangement of atoms (groups) in space



Isomerism

Structural isomerism

(Constitutional isomers-
The same molecular formula,
but different bonding
arrangement among the
atoms)

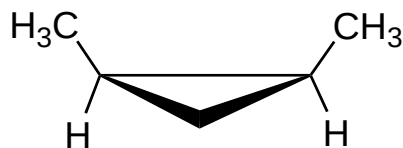
- i. Chain or nuclear isomers
- ii. Position isomers
- iii. Functional isomers

Stereoisomerism:

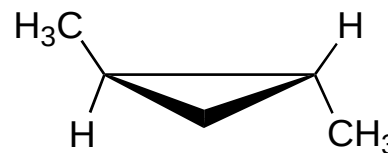
Optical isomerism

Geometrical isomerism (cis-trans)

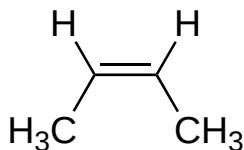
Stereoisomers: molecules with the same connectivity but different arrangement of atoms (groups) in space



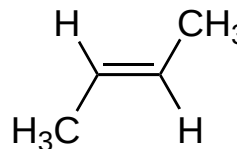
cis-1,2-dimethylcyclopropane



trans-1,2-dimethylcyclopropane



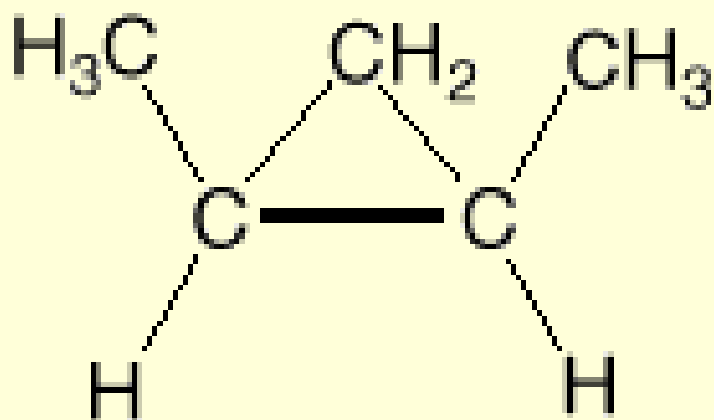
cis-2-butene



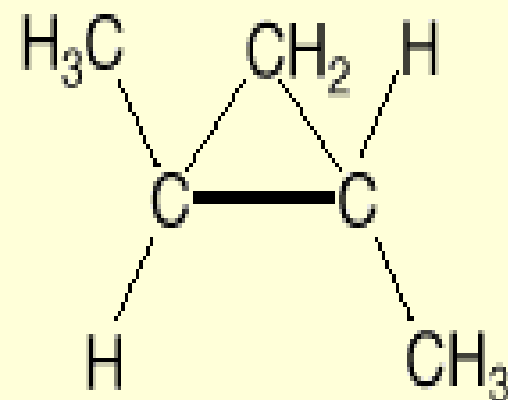
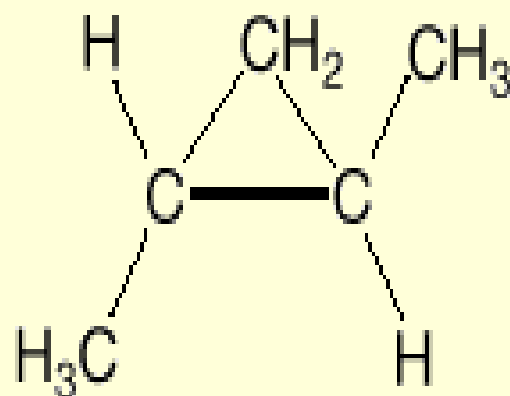
trans-2-butene

geometric isomers (diastereomers)

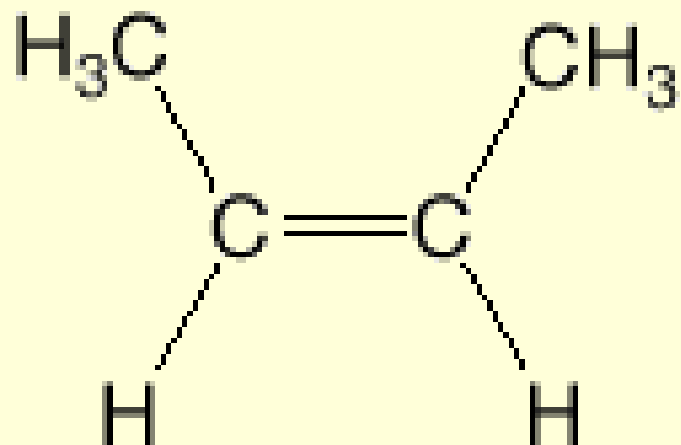
Geometrical Isomerism in Cyclic Systems (cis-trans)



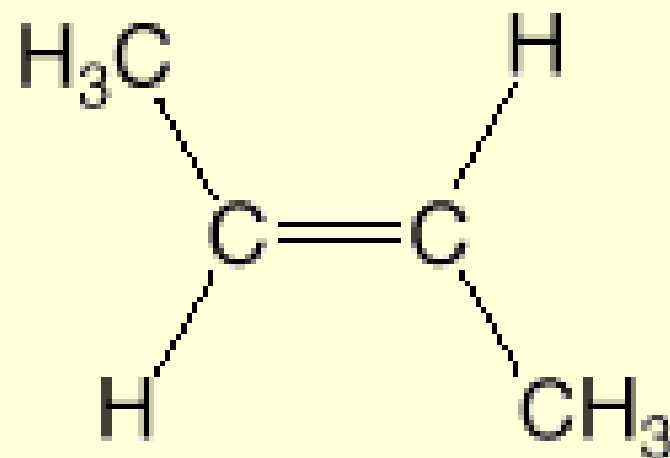
***cis*-1,2-Dimethylcyclopropane**



Two representations of
***trans*-1,2-dimethylcyclopropane**



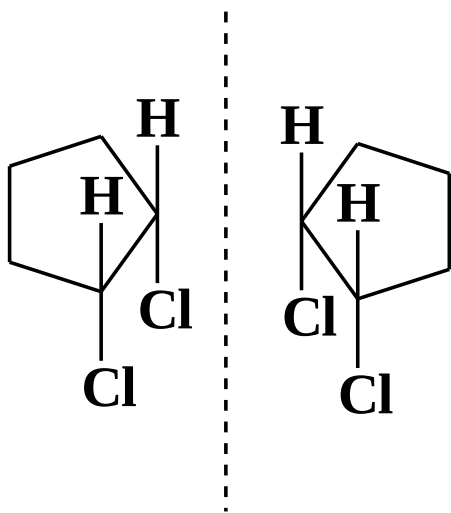
***cis*-2-Butene [or (*Z*)-2-butene]**



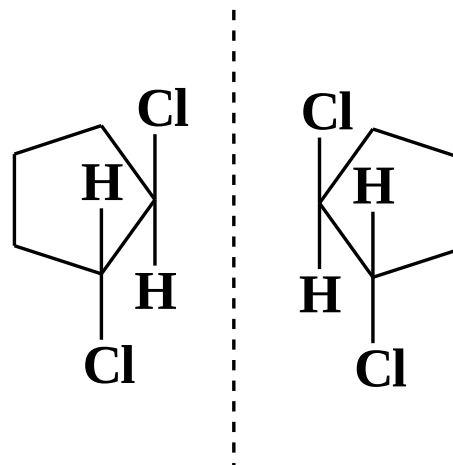
***trans*-2-Butene [or (*E*)-2-butene]**

Stereoisomers

- **Geometric** isomers: cis-trans isomers.
- **Enantiomers**: nonsuperimposable mirror images, different molecules.



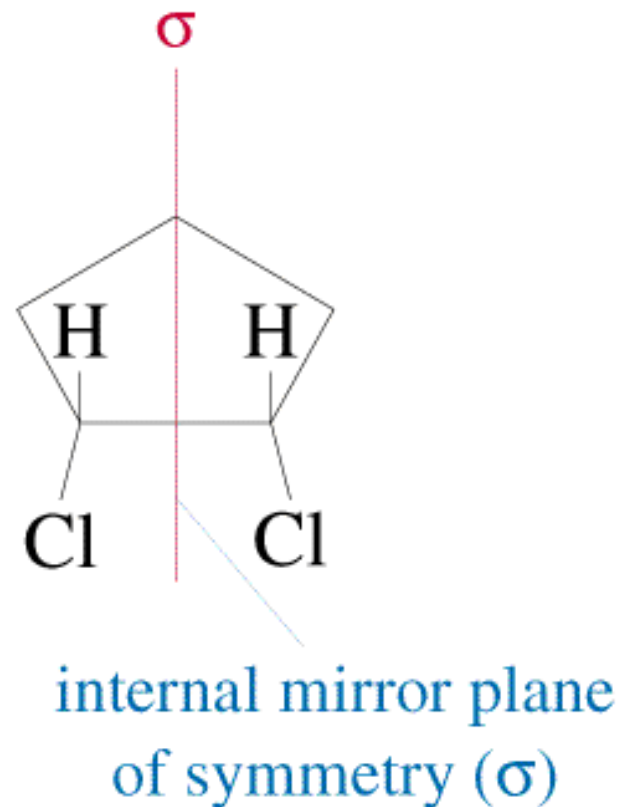
cis-1,2-dichlorocyclopentane



trans -1,2-dichlorocyclopentane

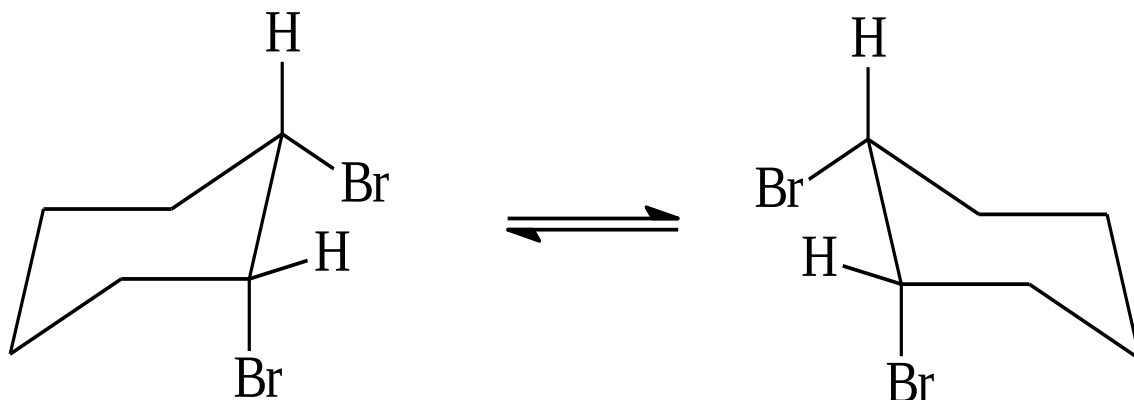
Mirror Planes of Symmetry

- If two groups are the same, carbon is **achiral**.
(animation)
- A molecule with an internal mirror plane cannot be chiral.*

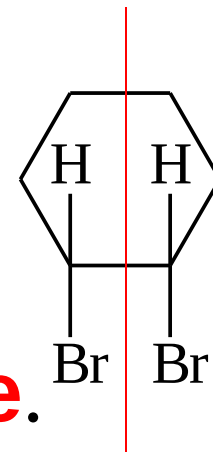


Caution! If there is no plane of symmetry, molecule may be chiral or achiral. See if mirror image can be superimposed. =>

Mobile Conformers



Nonsuperimposable mirror images, but equal energy and **interconvertible**.

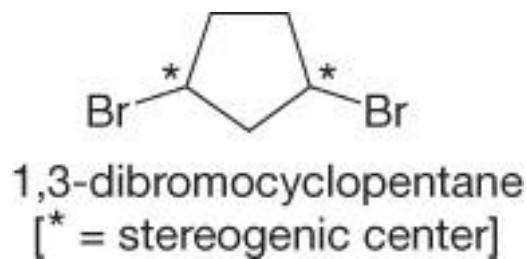


Use planar approximation

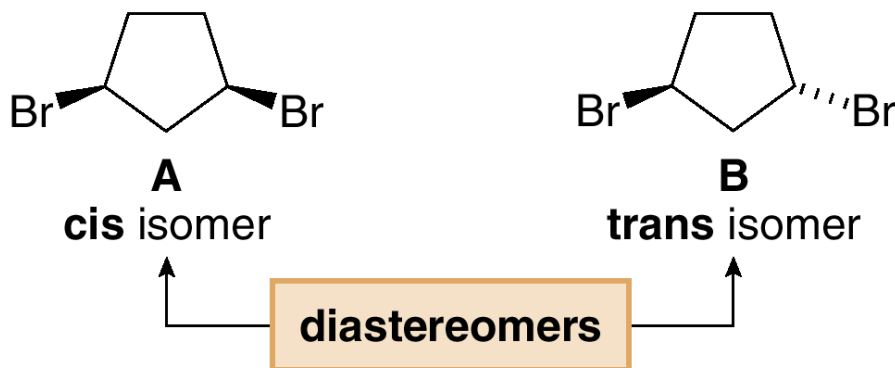
Stereochemistry

Disubstituted Cycloalkanes:

- Consider 1,3-dibromocyclopentane. Since it has two stereogenic centers, it has a maximum of four stereoisomers.



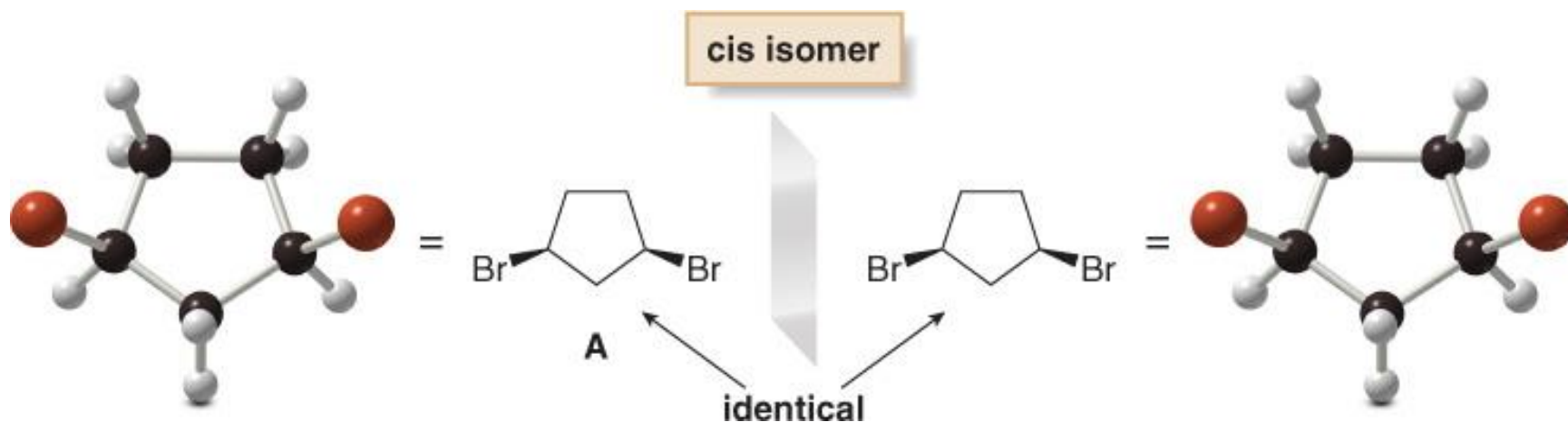
- Recall that a disubstituted cycloalkane can have two substituents on the same side of the ring (*cis* isomer, A) or on opposite sides of the ring (*trans* isomer, B). These compounds are stereoisomers but not mirror images.



Stereochemistry

Disubstituted Cycloalkanes:

- To find the other two stereoisomers if they exist, draw the mirror images of each compound and determine whether the compound and its mirror image are superimposable.

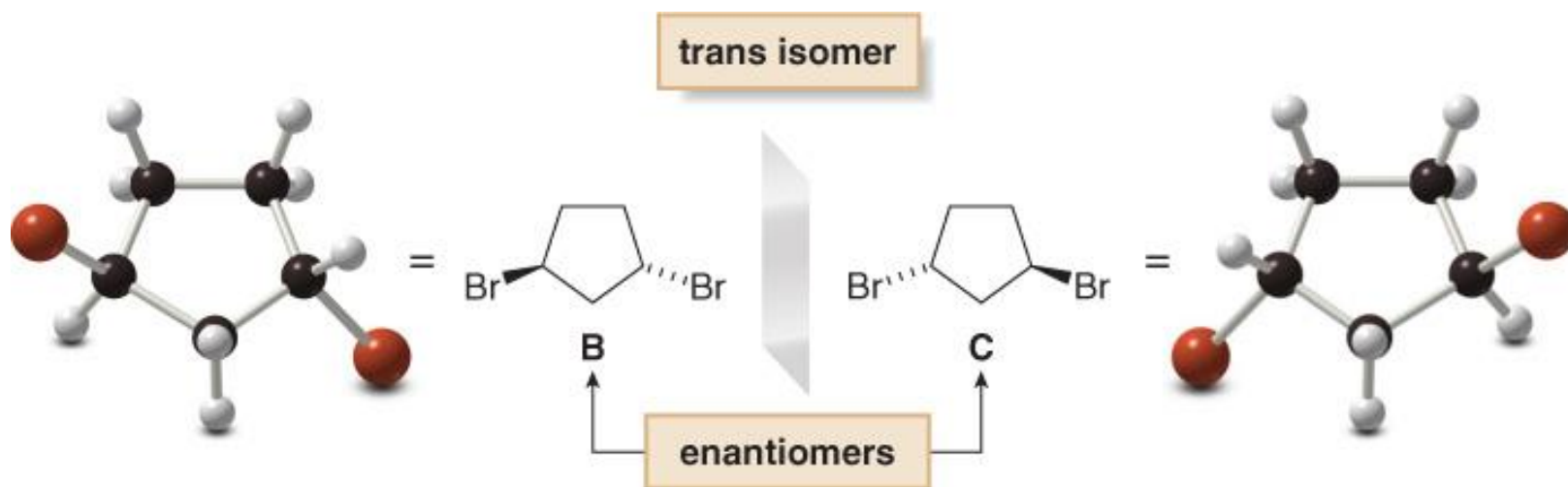


- The *cis* isomer is superimposable on its mirror image, making the images identical. Thus, A is an achiral **meso compound**.

Stereochemistry

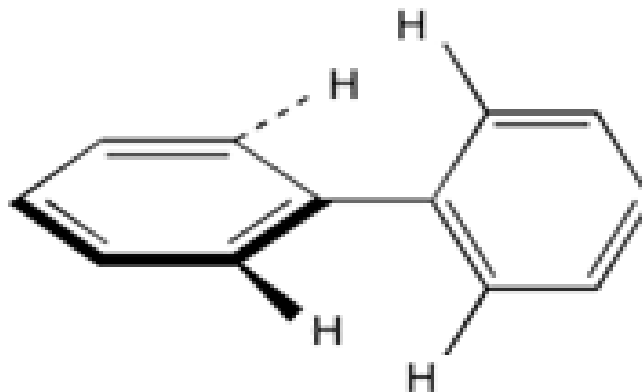
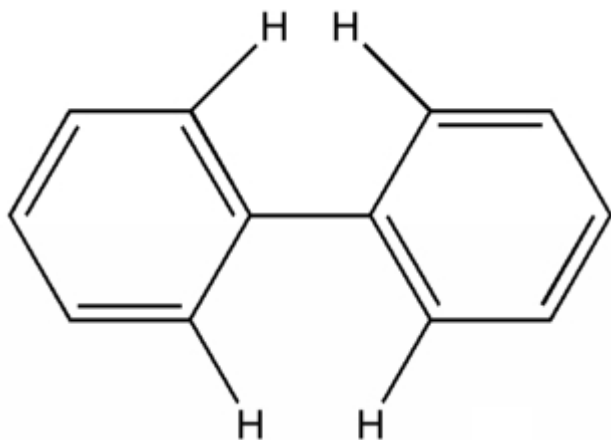
Disubstituted Cycloalkanes:

- The trans isomer is not superimposable on its mirror image, labeled C, making B and C different compounds. B and C are enantiomers.



- Because one stereoisomer of 1,3-dibromocyclopentane is superimposable on its mirror image, there are only three stereoisomers, not four.

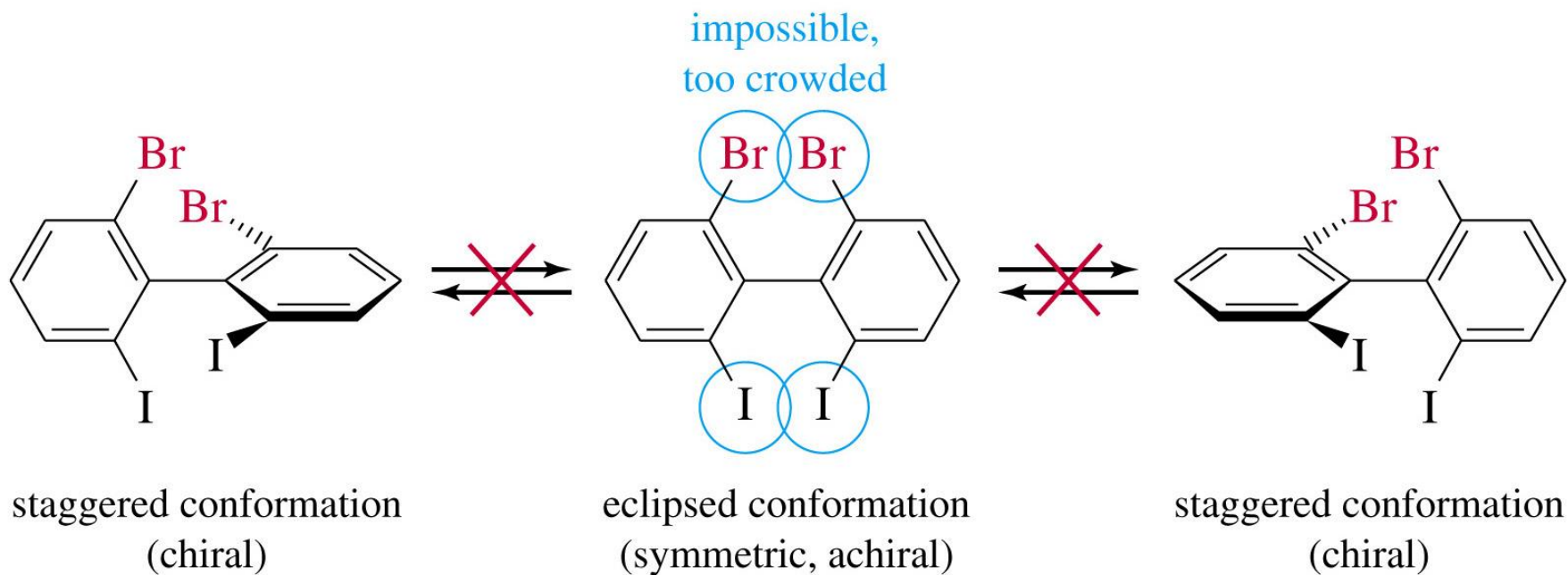
- The ortho – hydrogens of the biphenyl molecules are closer together than the sum of their Vander Waals radius. Therefore two benzene rings are perpendicular to each other naturally.



However, the steric interaction of these hydrogen atoms are not enough to prevent the free rotation completely.

The two conformational enantiomers of the biphenyl molecule can be isolated individually.

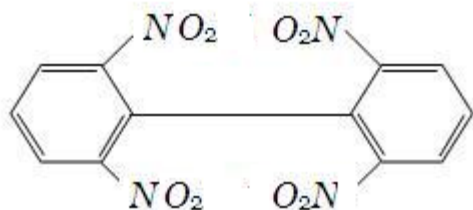
If the conformer is sterically hindered, it may exist as enantiomers, thus optically active



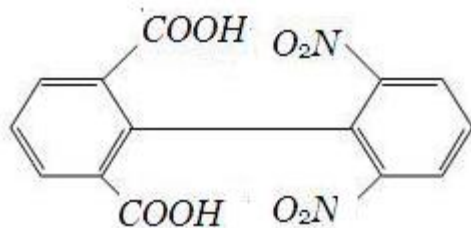
Such types of stereoisomerism which is due to restricted rotation about single bond, is known as atropisomerism and the stereoisomers are known as atropisomers.



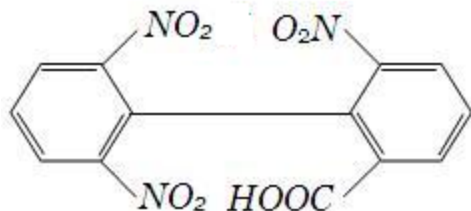
Optically inactive



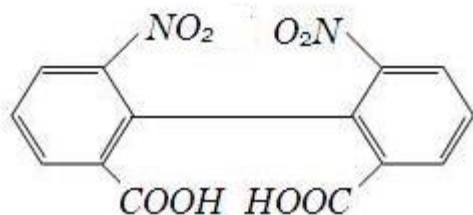
Optically inactive (Plane of symmetry)



Optically inactive (Plane of symmetry)



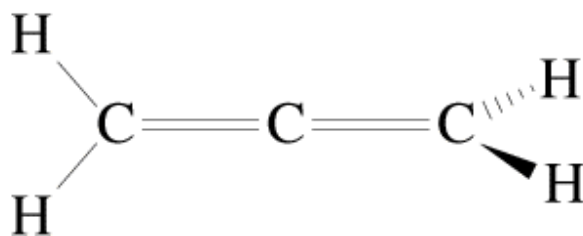
Optically inactive (Plane of symmetry in ring 1)



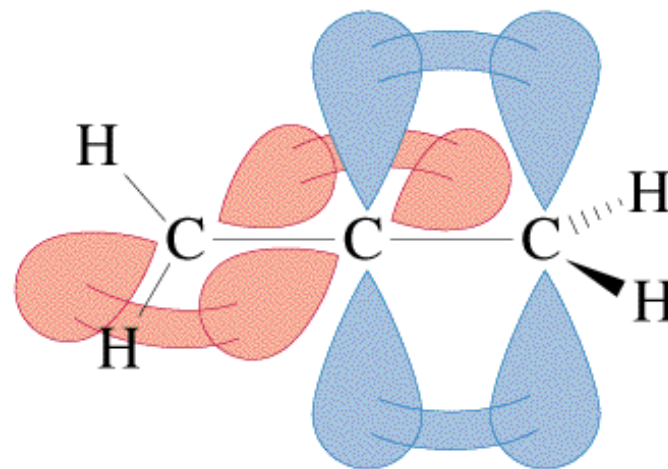
Optically active (Absence of plane of symmetry)

Allenes

- Some allenes also show chirality in the absence of asymmetric carbon atoms.
- Contains sp hybridized carbon with adjacent double bonds: $-C=C=C-$
- End carbons must have different groups.



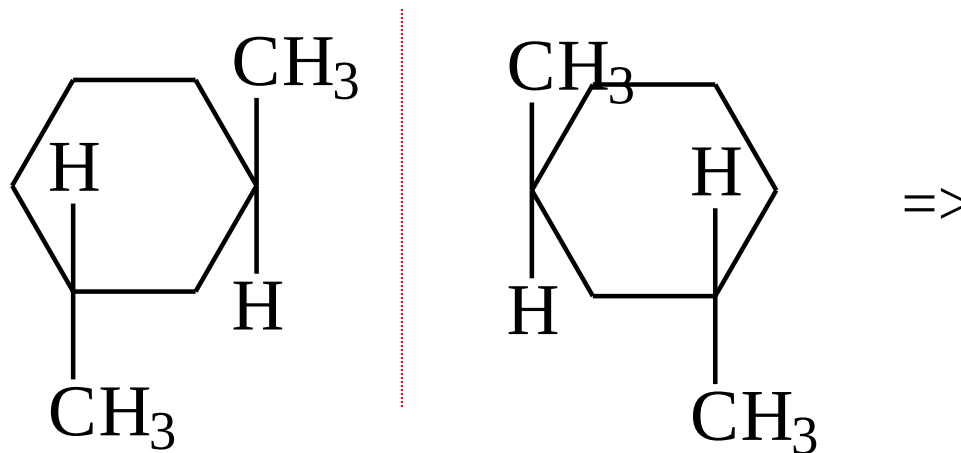
Allene is achiral.



=>

Ring Compounds

- Cis-trans isomers possible.
- May also have enantiomers.
- Example: *trans*-1,3-dimethylcyclohexane



Properties of Diastereomers

- Diastereomers have different physical properties: m.p., b.p.
- They can be separated easily.
- Enantiomers differ only in reaction with other chiral molecules and the direction in which polarized light is rotated.
- Enantiomers are difficult to separate.

Chromatographic Resolution of Enantiomers

